Glyphosate Exposure Analysis

A Comprehensive Study of Biomarker Responses to Herbicide Treatment

Statistical Analysis | Oxidative Stress Assessment | Inflammatory Response

Executive Summary

Key findings from the biomarker analysis

Study Design

- **Control Group:** Saline only
- **Low Dose:** 10% glyphosate in saline
- **Medium Dose:** 20% glyphosate in saline
- **High Dose:** 30% glyphosate in saline

Key Findings

- ✓ No statistically significant differences in ANOVA ($p > 0.05$)
- ✓ Large effect sizes for Catalase and Total Protein
- ✓ Kruskal-Wallis confirmed ANOVA results
- ✓ Potential biological effects despite statistical insignificance

Biomarkers Evaluated

Six key indicators of oxidative stress and inflammation

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Catalase

U/mg protein

Antioxidant enzyme

Total Protein

g/L

Protein synthesis marker

SOD

U/mg protein

Superoxide dismutase

GSH

mM

Reduced glutathione

MDA

µM

Lipid peroxidation marker

IL-1β

pg/ml

Inflammatory cytokine

Descriptive Statistics

Mean values and standard deviations across treatment groups

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Parameter	Control	Low Dose	Medium Dose	High Dose
Catalase (U/mg protein)	9.07 ± 6.54	22.25 ± 8.37	24.92 ± 12.63	19.30 ± 12.06
Total Protein (g/L)	12.70 ± 2.99	9.11 ± 2.64	8.53 ± 2.19	9.48 ± 1.72
SOD (U/mg protein)	0.0597 ± 0.018	0.0793 ± 0.031	0.0678 ± 0.032	0.1093 ± 0.097
GSH (mM)	0.754 ± 0.118	0.632 ± 0.052	0.592 ± 0.186	0.812 ± 0.254
MDA (µM)	0.564 ± 0.043	0.602 ± 0.064	0.548 ± 0.043	0.548 ± 0.053
IL-1β (pg/ml)	533.27 ± 119.78	581.46 ± 99.40	522.80 ± 44.77	512.54 ± 52.13

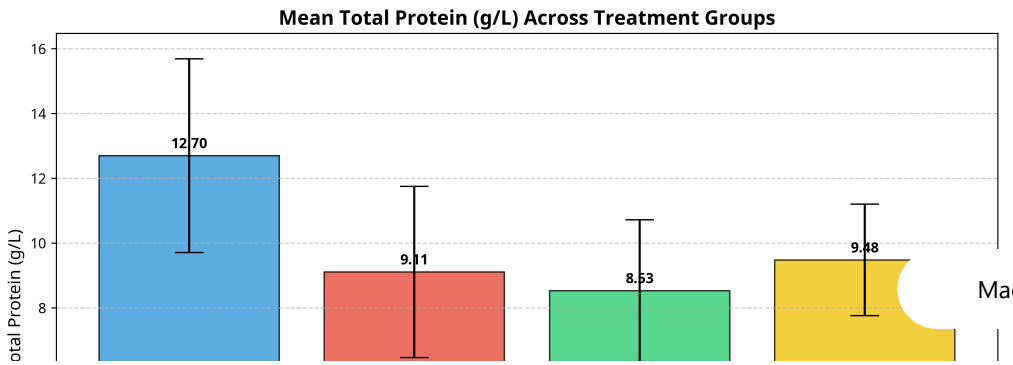
Data Visualizations

Mean biomarker concentrations across treatment groups with error bars

Catalase | Protein | SOD

GSH | MDA | IL-1β

Total Protein



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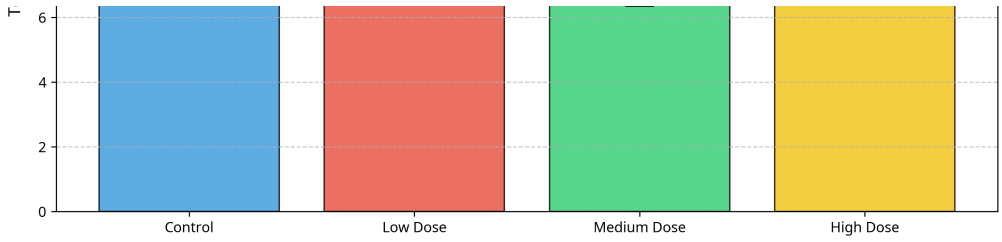

Statistical Analysis

Assumption testing and inferential statistics

ANOVA Results | Kruskal-Wallis | Effect Size

One-way ANOVA was performed to compare biomarker levels across treatment groups. All results showed $p > 0.05$, indicating no statistically significant differences.

Parameter	F value	p-value	Result
Catalase	1.475	0.271	Not Significant
Total Protein	2.553	0.104	Not Significant
SOD	0.201	0.894	Not Significant
GSH	1.101	0.387	Not Significant
MDA	1.966	0.173	Not Significant
IL-1β	0.291	0.831	Not Significant

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Methodology & Assumptions

Key Results & Interpretation

Discussion & Conclusions

Study Methodology

This analysis employed rigorous statistical methods including descriptive statistics, assumption testing (Shapiro-Wilk and Levene's tests), one-way ANOVA, Kruskal-Wallis non-parametric analysis, and effect size calculations using Cohen's d. All analyses were performed using SPSS statistical software.

Data represents mean values ± standard deviations from experimental groups. Statistical significance threshold: $\alpha = 0.05$. Effect size interpretation: small ($d = 0.2$), medium ($d = 0.5$), large ($d \geq 0.8$).

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